

Applied Bayesian Statistics in Animal & Biological Sciences

4) Non-linear regression and heteroscedasticity

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Today's program

Methodology

- Bayesian non-linear regression
- Heteroscedasticity and model selection
- Posterior predictive distribution

Today's program

Methodology

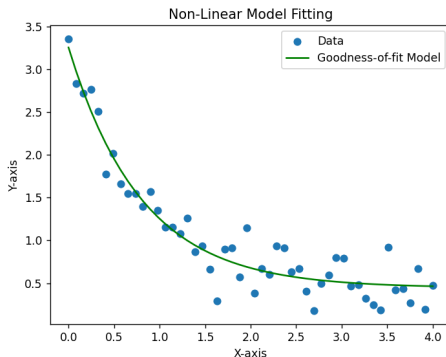
- Bayesian non-linear regression
- Heteroscedasticity and model selection
- Posterior predictive distribution

Toxicokinetic modeling datasets

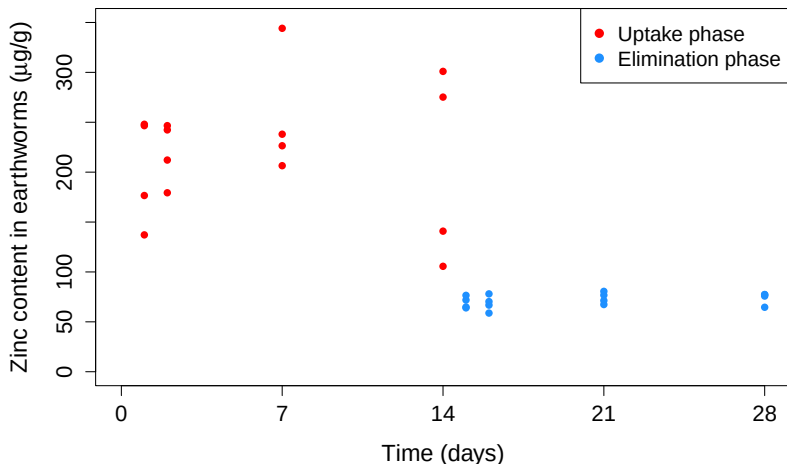
- Zinc and manganese in *Eisenia fetida* (earthworms) exposed to nanomaterials
- Propranolol in *Gammarus Pulex* (Freshwater shrimp/amphipod crustaceans) exposed to polluted water

General and Bayesian non-linear regression

- Non-linearity in parameters with respect to the response variable, e.g. $y = a + x^b + \epsilon$; ($y = a + bx + cx^2 + \epsilon$ is a linear modeling)
- Parameters may have a specific biological meaning
- Non-linear models may be more parsimonious (i.e. fewer parameters) than a competitor linear model, such as a polynomial
- Bayesian non-linear modeling uses priors for all model parameters



Ecotoxicological trial - earthworms exposed to ZnO:Mn



- Worms in contaminated soils (uptake) and clean soils (elimination)
- Different variances in Zn contents in the different phases

Physiology-based toxicokinetic modeling

Uptake phase: $0 < t \leq t_n$

$$C(t, t_n, C_{\text{exp}}; \boldsymbol{\theta}) = C_0 + \frac{k_1}{k_2} C_{\text{exp}} (1 - e^{-k_2 t}) \text{ with } \boldsymbol{\theta} = [C_0, k_1, k_2] \quad (1)$$

Physiology-based toxicokinetic modeling

Uptake phase: $0 < t \leq t_n$

$$C(t, t_n, C_{\text{exp}}; \boldsymbol{\theta}) = C_0 + \frac{k_1}{k_2} C_{\text{exp}} (1 - e^{-k_2 t}) \text{ with } \boldsymbol{\theta} = [C_0, k_1, k_2] \quad (1)$$

Elimination phase: $t > t_n$

$$C(t, t_n, C_{\text{exp}}; \boldsymbol{\theta}) = C_0 + \frac{k_1}{k_2} C_{\text{exp}} (e^{k_2(t-t_n)} - e^{-k_2 t}) \quad (2)$$

Physiology-based toxicokinetic modeling

Uptake phase: $0 < t \leq t_n$

$$C(t, t_n, C_{\text{exp}}; \boldsymbol{\theta}) = C_0 + \frac{k_1}{k_2} C_{\text{exp}} (1 - e^{-k_2 t}) \text{ with } \boldsymbol{\theta} = [C_0, k_1, k_2] \quad (1)$$

Elimination phase: $t > t_n$

$$C(t, t_n, C_{\text{exp}}; \boldsymbol{\theta}) = C_0 + \frac{k_1}{k_2} C_{\text{exp}} (e^{k_2(t-t_n)} - e^{-k_2 t}) \quad (2)$$

Equal or unequal variance along uptake and elimination phases?

$$y_{ij} = C(t, \dots) + e_{ij} \text{ with } e_{ij} \sim N(0, \sigma_e^2) \quad (3)$$

$$y_{ijk} = C(t, \dots) + e_{ijk} \text{ with } e_{ijk} \sim N(0, \delta_k \sigma_e^2) \quad (4)$$

Entity i at time j in phase k

Physiology-based toxicokinetic modeling

Uptake phase: $0 < t \leq t_n$

$$C(t, t_n, C_{\text{exp}}; \boldsymbol{\theta}) = C_0 + \frac{k_1}{k_2} C_{\text{exp}} (1 - e^{-k_2 t}) \text{ with } \boldsymbol{\theta} = [C_0, k_1, k_2] \quad (1)$$

Elimination phase: $t > t_n$

$$C(t, t_n, C_{\text{exp}}; \boldsymbol{\theta}) = C_0 + \frac{k_1}{k_2} C_{\text{exp}} (e^{k_2(t-t_n)} - e^{-k_2 t}) \quad (2)$$

Equal or unequal variance along uptake and elimination phases?

$$y_{ij} = C(t, \dots) + e_{ij} \text{ with } e_{ij} \sim N(0, \sigma_e^2) \quad (3)$$

$$y_{ijk} = C(t, \dots) + e_{ijk} \text{ with } e_{ijk} \sim N(0, \delta_k \sigma_e^2) \quad (4)$$

Entity i at time j in phase k

- k_1 and k_2 represent the uptake and elimination rates of the chemical

Bayesian non-linear regression in STAN - 1

```
mod_NLin_ToxKin <- "  
data {  
  int<lower=0> N; // Number of observations  
  real x[N]; // time  
  real y[N]; // Zn concentration in earthworms  
  real z; int t_d; // Zn concentration in soil and soil transfer time  
  int grp[N]; int no_sigma; // Sigma stratification variable and number of sigma's  
  
  real a_C0; real b_C0; // Prior parameters for C_0  
  real mean_k1; real sd_k1; // Prior parameters for k_1  
  real mean_k2; real sd_k2; // Prior parameters for k_2  
} // End of data block  
  
parameters {  
  real<lower=0> C_0; real<lower=0> k[2]; // Contains C0, k1 and k2.  
  real<lower=0> sigma_e[no_sigma]; // residual sd  
} // End of parameters block  
  
transformed parameters {  
  real m[N];  
  
  for (i in 1:N){  
    m[i] = (x[i] <= t_d ? k[1]/k[2]*z*(1-exp(-k[2]*x[i])) :  
            k[1]/k[2]*z*(exp(-k[2]*(x[i]-t_d))-exp(-k[2]*x[i])));  
  } // End of for loop i; note the meaning of '?' and ':'  
} // End of transformed parameters block with one-compartmental function  
  
. . .
```

Bayesian non-linear regression in STAN - 2

```
...
model {
  // priors
  C_0 ~ normal(a_C0, b_C0); // Assign a gamma prior to C_0
  k[1] ~ normal(mean_k1, sd_k1); k[2] ~ normal(mean_k2, sd_k2); // Assign normal priors to k1 and k2
  sigma_e ~ cauchy(0,5); // Assign a half-cauchy prior to residual variance parameter sigma

  // likelihoods
  for (j in 1:N) y[j] ~ normal(C_0 + m[j], sigma_e[grp[j]]); // Toxkin model
} // End of model block

generated quantities {
  // Initialize (additional) quantities to be generated
  vector[N] log_lik; // to use loo package and compute waic - the logLikelihood MUST be called log_lik!
  real pred[N]; // Quantities for fitted values (not including noise/error estimate)
  real res[N]; // Residuals

  // Assign values to the initialized quantities to be returned
  for (n in 1:N){
    pred[n] = C_0 + (x[n] <= t_d ? k[1]/k[2]*z*(1-exp(-k[2]*x[n])) :
                                                              k[1]/k[2]*z*(exp(-k[2]*(x[n]-t_d))-exp(-k[2]*x[n])));

    res[n] = y[n] - pred[n];

    log_lik[n] = normal_lpdf(y[n] | pred[n], sigma_e[grp[n]]); // Toxkin model
  } // End of for loop n
} // End of generated quantities block
"
```

Bayesian non-linear regression in RSTAN - 1

```
library(rstan)
options(mc.cores = parallel::detectCores())
rstan_options(auto_write = TRUE)
library(loo)

### Compile stan model
stan_mod_NLin-ToxKin <- stan_model(model_code=mod_NLin-ToxKin, model_name="mod_NLin-ToxKin")

### Read and process data
worm_Zn_dat <- read.table("Efetida_Zn.txt", header=T, sep=",")
worm_Zn_dat <- data.frame(worm_Zn_dat, Phase=1+as.integer(worm_Zn_dat$Time > 14))

worm_dat.list <- list(N=nrow(worm_Zn_dat), # Number of observations
                      x=worm_Zn_dat$Time, # time
                      y=worm_Zn_dat$toxicant, # Zn concentration in earthworms
                      z=mean(worm_Zn_dat$C_exp), # Zn conc in soil
                      t_d=14, # soil transfer time
                      grp=rep(1, worm_Zn_dat), # Sigma stratification variable
                      no_sigma=1, # number of sigma's
                      a_C0=75, b_C0=5, # Prior parameters for C_0
                      mean_k1=1, sd_k1=1, # Prior parameters for k_1
                      mean_k2=1, sd_k2=1) # Prior parameters for k_2

worm_Zn_homosc.rs <- sampling(stan_mod_NLin-ToxKin, data=worm_dat.list, iter=30e3,
                             pars=c("C_0", "k", "sigma_e", "res", "log_lik"), warmup=1e3, thin=25, chains=2)

worm_dat.list$grp <- worm_dat.list$Phase
worm_dat.list$no_sigma <- 2
worm_Zn_heterosc.rs <- sampling(stan_mod_NLin-ToxKin, data=worm_dat.list, iter=30e3,
                                pars=c("C_0", "k", "sigma_e", "res", "log_lik"), warmup=1e3, thin=25, chains=2)
```

Bayesian non-linear regression in RSTAN - 2

```
print(summary(worm_Zn_homosc.rs, pars=c("C_0","k","sigma_e"))$summary)
```

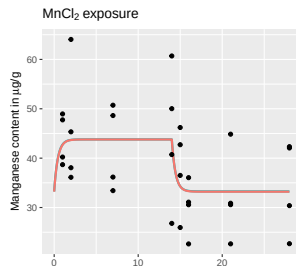
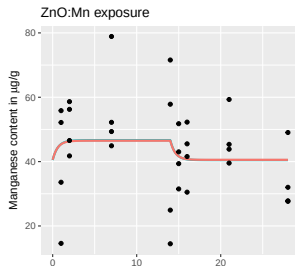
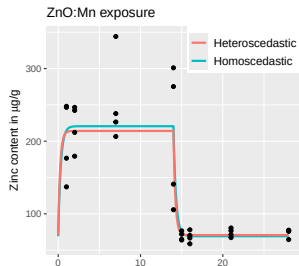
##	mean	se_mean	sd	2.5%	25%	50%	75%	97.5%	n_eff	Rhat
## C_0	73.645	0.170	4.794	64.431	70.171	73.578	76.815	83.035	795.424	1.000
## k[1]	0.530	0.005	0.129	0.318	0.435	0.517	0.608	0.812	715.261	1.000
## k[2]	2.405	0.022	0.596	1.471	1.975	2.323	2.761	3.736	719.660	0.999
## sigma_e[1]	44.371	0.215	5.801	34.646	40.204	43.923	47.880	57.740	727.920	0.999

```
print(summary(worm_Zn_heterosc.rs, pars=c("C_0","k","sigma_e"))$summary)
```

##	mean	se_mean	sd	2.5%	25%	50%	75%	97.5%	n_eff	Rhat
## C_0	70.806	0.034	1.832	67.130	69.639	70.827	72.030	74.326	2932.903	1.000
## k[1]	0.783	0.003	0.147	0.517	0.681	0.772	0.878	1.105	3231.801	0.999
## k[2]	3.614	0.009	0.519	2.712	3.241	3.570	3.938	4.756	3150.903	0.999
## sigma_e[1]	63.087	0.218	12.032	44.740	54.608	61.297	69.607	91.601	3039.504	1.000
## sigma_e[2]	7.370	0.027	1.474	5.145	6.309	7.161	8.196	10.846	2923.347	1.000

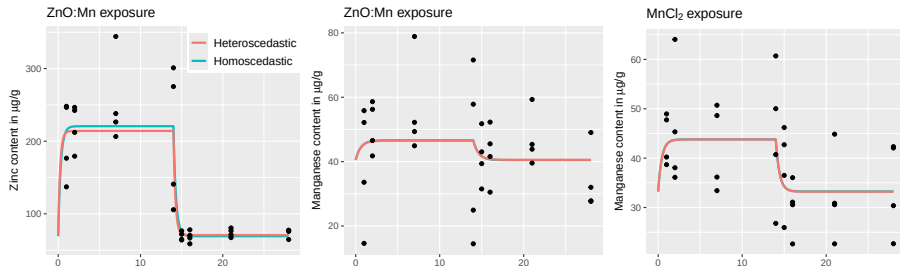
- Note the $\sigma_{e,1}$ and $\sigma_{e,2}$ values for the heteroscedastic model

Bayesian non-linear prediction of Zn in earthworms



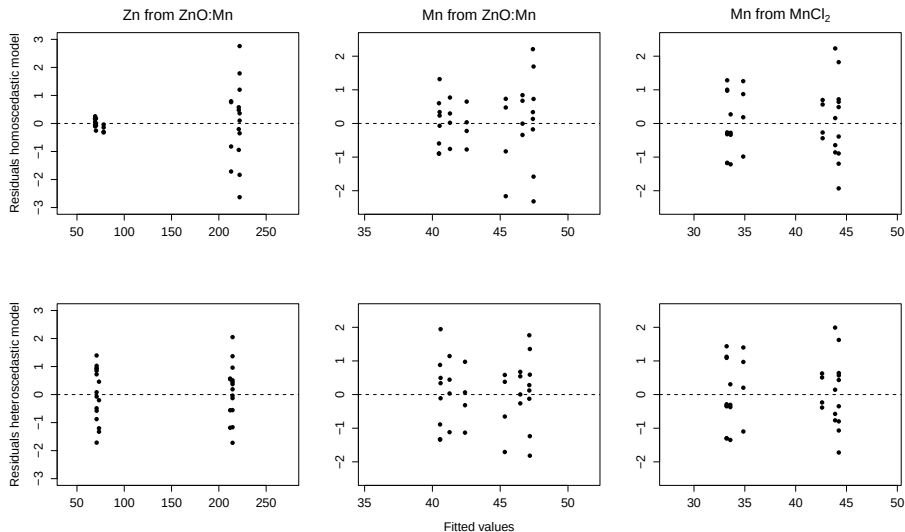
Model fit	C_0	k_1	k_2	σ_e or $\sigma_{e,\text{uptake}}$	$\sigma_{e,\text{elimin}}$
Homoscedastic Zn	73.6 ± 4.8	0.530 ± 0.129	2.41 ± 0.60	44.4 ± 5.8	
Heteroscedastic Zn	70.8 ± 1.8	0.783 ± 0.147	3.61 ± 0.52	63.1 ± 12.0	7.4 ± 1.5

Bayesian non-linear prediction of Zn in earthworms



Model fit	C_0	k_1	k_2	σ_e or $\sigma_{e,\text{uptake}}$	$\sigma_{e,\text{elimin}}$
Homoscedastic Zn	73.6±4.8	0.530±0.129	2.41±0.60	44.4±5.8	
Heteroscedastic Zn	70.8±1.8	0.783±0.147	3.61±0.52	63.1±12.0	7.4±1.5
Homoscedastic Mn	40.5±2.3	0.0515±0.0296	1.64±0.81	14.2±1.8	
Heteroscedastic Mn	40.6±2.0	0.0508±0.0306	1.67±0.80	18.0±3.4	9.62±1.81
Homoscedastic Mn	33.2±1.7	0.115±0.041	2.31±0.84	9.04±1.18	
Heteroscedastic Mn	33.2±1.6	0.114±0.042	2.27±0.85	10.1±1.9	8.11±1.51

Bayesian non-linear residual variance modeling - residuals



Bayesian non-linear regression in RSTAN - 3

```
compute_loo <- function(stan_out.rs){  
  # Extract log likelihood; merge_chains=T indicates log-likelihoods merge across chains  
  loglik_mod.rs <- extract_log_lik(stan_out.rs, parameter_name="log_lik", merge_chains=T)  
  
  LL <- as.array(stan_out.rs, pars="log_lik" # Convert log Likelihood into an array  
  
  # Compute rel. efficiency of the chains and then looic using log-likelihood and r_eff  
  r_eff <- relative_eff(exp(LL), cores=1)  
  ltest <- loo(loglik_mod.rs, r_eff=r_eff, cores=1)  
  return(ltest) # Return LOOIC  
} # End of compute_loo()
```

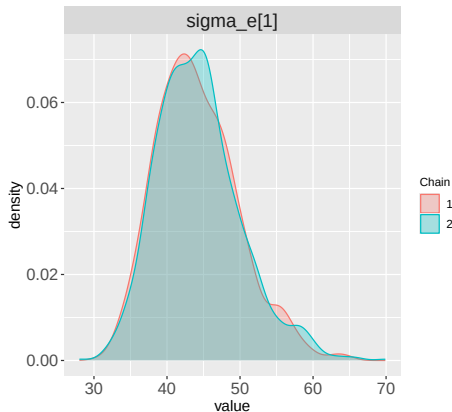
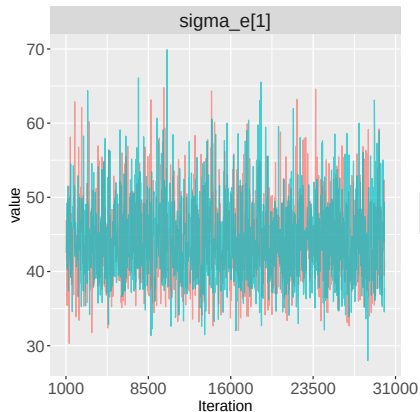
```
homosc.loo <- compute_loo(worm_Zn_homosc.rs)  
heterosc.loo <- compute_loo(worm_Zn_heterosc.rs)
```

```
print(homosc.loo$estimates)  
##           Estimate           SE  
## elpd_loo -169.4749   6.719187  
## looic      338.9498  13.438374
```

```
print(heterosc.loo$estimates)  
##           Estimate           SE  
## elpd_loo -145.834374   7.2614507  
## looic      291.668747  14.5229014
```

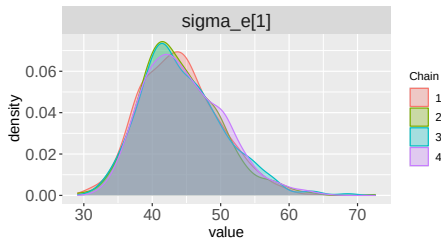
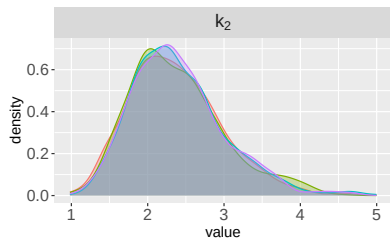
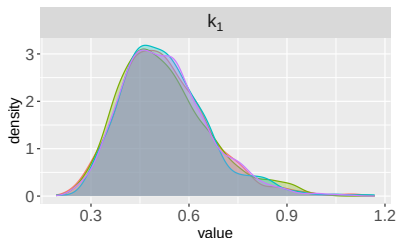
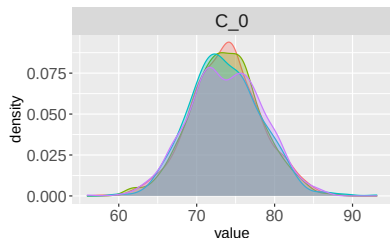
- LOOIC obviously favors the heteroscedastic model

Traceplots and density plots of model parameters



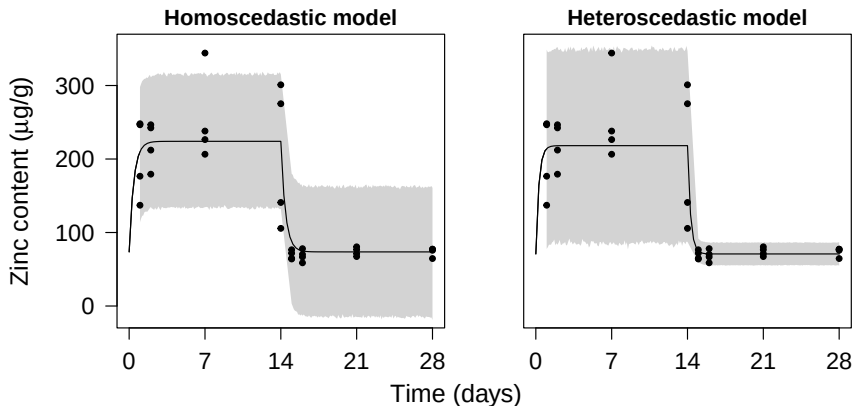
- Both plots visualize chain convergence, density plot visualizes the parameter distribution the best

Posterior (predictive) distribution



- Predict future values of the response variable: $p(\tilde{y}|\mathbf{y}) = \int p(\tilde{y}|\theta)p(\theta|\mathbf{y})d\theta$
- Integrate whole parameter space out

Posterior predictive distribution for Zn content in *E. fetida*



- Note the different credible interval widths in the uptake and elimination phases for the heteroscedastic model vs the homoscedastic model

Posterior predictive distribution in STAN - 1

```
mod_NLin_ToxKin02 <- "
data {
  int<lower=0> N; // Number of observations
  real x[N]; // time
  real y[N]; // Zn concentration in earthworms
  real z; int t_d; // Zn concentration in soil and soil transfer time
  int grp[N]; int no_sigma; // Sigma stratification variable and number of sigma's
  int<lower=N> P; real pred_t[P]; int s_grp[P]; // ...

  real a_C0; real b_C0; // Prior parameters for C_0
  real mean_k1; real sd_k1; // Prior parameters for k_1
  real mean_k2; real sd_k2; // Prior parameters for k_2
} // End of data block
```

- Posterior predictive distribution is simulated for P time points (`pred_t`) throughout uptake and elimination phases (`s_grp`)
- Parameter, transformed parameter and model blocks unchanged
- Code for Posterior predictive distribution in the generated quantities block...

Posterior predictive distribution in STAN - 2

```
generated quantities {  
  // to use loo package and compute waic, the logLikelihood MUST be named log_lik!  
  vector[N] log_lik;  
  
  // Initialize (additional) quantities to be generated  
  real pred[N]; // Quantities for fitted values (not including noise/error estimate)  
  real res[N];  // Residuals  
  real cpred[P]; real ypred[P]; // Predicted values (including noise/error estimate)  
  
  // Assign values to the initialized quantities to be returned  
  for (n in 1:N){  
    pred[n] = C_0 + (x[n] <= t_d ? k[1]/k[2]*z*(1-exp(-k[2]*x[n])) :  
                                                             k[1]/k[2]*z*(exp(-k[2]*(x[n]-t_d))-exp(-k[2]*x[n])));  
    res[n] = y[n] - pred[n];  
  
    log_lik[n] = normal_lpdf(y[n] | pred[n], sigma_e[grp[n]]); // Toxkin model  
  
  } // End of for loop n  
  
  for (p in 1:P){  
    cpred[p] = (pred_t[p] <= t_d ? k[1]/k[2]*z*(1-exp(-k[2]*pred_t[p])) :  
                                                         k[1]/k[2]*z*(exp(-k[2]*(pred_t[p]-t_d))-exp(-k[2]*pred_t[p])));  
  
    ypred[p] = normal_rng(C_0 + cpred[p], sigma_e[s_grp[p]]); // Toxkin model  
  
  } // End of for loop p  
// End of generated quantities block  
}
```

Posterior predictive distribution RSTAN - 4

```
# Compile stan model
stan_mod_NLin-ToxKin02 <- stan_model(model_code=mod_NLin-ToxKin02, model_name="mod_NLin-ToxKin02")

# Data points for random number generator that samples posterior predictive distribution
t_rng <- seq(1,14,0.1)
rng_grp <- rep(1,length(t_rng)); t_rng <- c(t_rng,14+t_rng); rng_grp <- c(rng_grp,rng_grp+1)

# Generate data list object and run simulations
worm_dat.list02 <- list(N=nrow(worm_Zn_dat), #Number of observations
  x=worm_Zn_dat$Time, # time
  y=worm_Zn_dat$toxicant, # Zn concentration in earthworms
  z=mean(worm_Zn_dat$C_exp), # Zn conc in soil
  t_d=14, # soil transfer time
  grp=rep(1,nrow(worm_Zn_dat)), # Sigma stratification variable
  no_sigma=1, # number of sigma's
  P=length(t_rng), pred_t=t_rng, s_grp=rng_grp,
  a_C0=75, b_C0=5, # Prior parameters for C_0
  mean_k1=1, sd_k1=1, # Prior parameters for k_1
  mean_k2=1, sd_k2=1) # Prior parameters for k_2

worm_Zn_homosc.rs02 <- sampling(stan_mod_NLin-ToxKin02, data=worm_dat.list02, iter=30e3, warmup=1e3,
  pars=c("C_0","k","sigma_e","log_lik","ypred"), thin=25, chains=2)

worm_dat.list02$grp<- worm_Zn_dat$Phase
worm_dat.list02$no_sigma <- 2
worm_Zn_heterosc.rs02 <- sampling(stan_mod_NLin-ToxKin02, data=worm_dat.list02, iter=30e3, warmup=1e3,
  pars=c("C_0","k","sigma_e","log_lik","ypred"), thin=25, chains=2)
```

- Generate timepoints from 1 to 14, then from 14 to 28; assign to t_rng

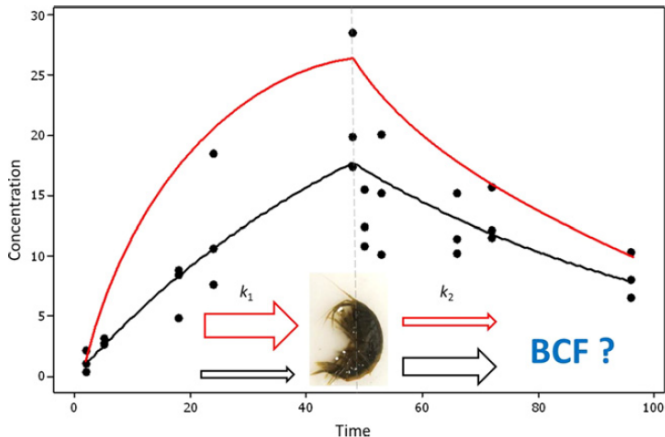
Posterior predictive distribution RSTAN - 5

```
print(summary(worm_Zn_homosc.rs02, pars=c("C_0", "k", "sigma_e", "ypred"))$summary)
```

##	mean	se_mean	sd	2.5%	25%	50%	75%	97.5%	r_eff	Rhat
##C_0	73.633	0.052	4.582	64.629	70.543	73.625	76.662	82.565	7796.106	1.000
##k[1]	0.526	0.001	0.132	0.311	0.432	0.511	0.606	0.826	8004.950	1.000
##k[2]	2.382	0.007	0.599	1.408	1.939	2.315	2.753	3.714	7967.616	1.000
##sigma_e[1]	44.364	0.065	5.828	34.784	40.220	43.793	47.862	57.248	7942.351	1.000
##ypred[1]	207.577	0.532	46.710	114.044	177.534	207.569	238.773	297.928	7696.155	1.000
##ypred[2]	210.418	0.524	46.989	118.990	178.908	210.376	241.578	302.134	8045.789	1.000

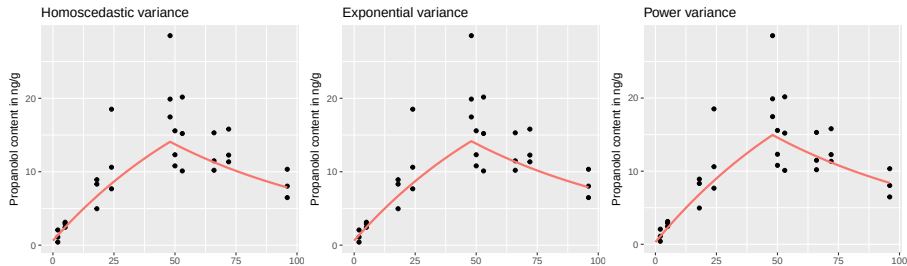
- Note 2.5% and 97.5% percentiles of `ypred` quantifying the bounds of the gray shaded area in the plot

Gammarus pulex toxicokinetic data example



Data from Miller et al. (2016) extracted using
<https://apps.automeris.io/wpd4/>

Gammarus pulex fits



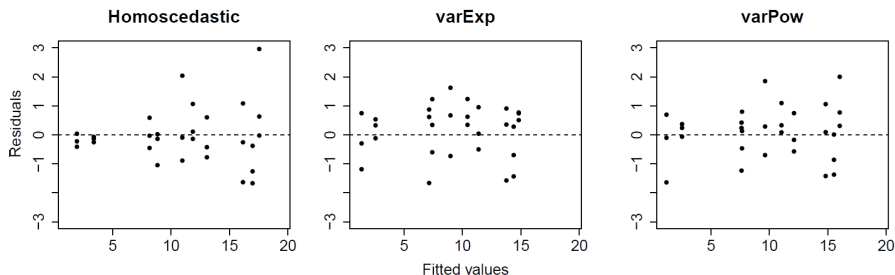
Model fit	C_0	k_1	k_2	σ_e	δ
Homoscedastic	0.922 ± 0.648	0.563 ± 0.054	0.0176 ± 0.0029	3.718 ± 0.513	
Exponential	0.527 ± 0.325	0.461 ± 0.043	0.0153 ± 0.0023	0.758 ± 0.258	0.111 ± 0.024
Power	0.259 ± 0.254	0.515 ± 0.047	0.0159 ± 0.0027	0.807 ± 0.318	0.612 ± 0.136

Homoscedastic, exponential or power variance, respectively:

$$e_{ij} \sim N(0, \sigma^2), e_{ij} \sim N(0, e^{2\delta y_{ij}} \sigma^2) \text{ or } e_{ij} \sim N(0, |y_{ij}|^{2\delta} \sigma^2) \quad (5)$$

- Credible interval of δ does not contain zero $\rightarrow$ heteroscedasticity

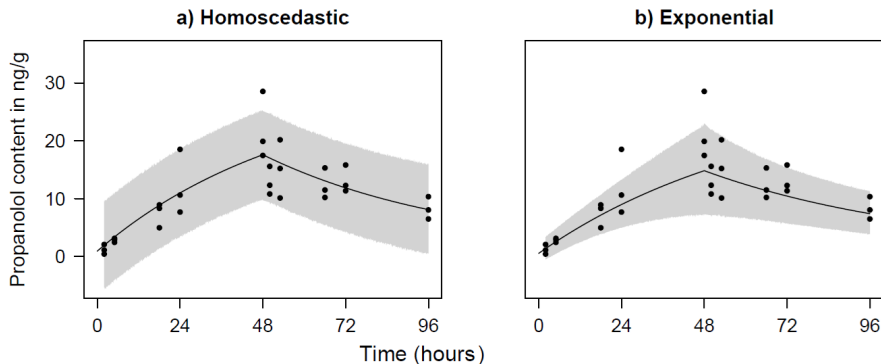
Residual plots of Bayesian non-linear regression



	Homoscedastic	var Exp	var Pow
LOOIC	167.6	139.7	149.6

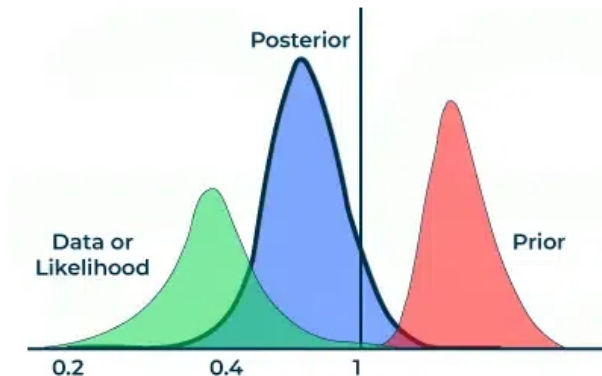
- Lowest LOOIC for exponential residual variance model

Posterior predictive distributions *G. pulex* models



- Exponential residual variance model follows data variance the best, but is not perfect with 2 points outside posterior predictive credible interval

Thank you for your attention!



Questions?

Exercise lecture 4

- Write STAN code for the *Gammarus pulex* data and use the various residual variance functions
- Insert as many `print()` statements as needed to check/understand what each line of code does
- Reconsider the priors and evaluate model simulation output